

CURRICULUM VITAE

RAHUL C KHOT

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PROFESSIONAL SUMMARY

Experienced **Embedded Hardware Design Engineer** with **2+ years** of expertise in circuit design & development, microcontroller interfaces, testing & debugging, PCB review, and technical documentation.

WORK EXPERIENCE (2Y+)

Organization	Designation	Period / Years of Exp
BDGtronix Pvt Ltd, Bengaluru	Embedded Hardware Design Engineer	From Jan 2024 to Till Date
ADM Agro Industries Latur & Vizag Pvt Ltd	Graduate Trainee- Engineer	July-2023 to Jan-2024

TECHNICAL SKILLS

- Practical experience in hardware design and development lifecycle.
- Analog (ADC) and Digital interface circuit designing and analysis.
- Low side and high side Output drivers circuit design and analysis.
- AC to DC, DC to DC, LDO, Switching Regulators design.
- Sensor Interfacing Circuit Design-Temperature, IR, Digital Switches and Speed.
- Output Interfacing Circuit Design-Relay, Motors, LEDs.
- Communication Interfaces UART, I2C, SPI and RS-422.
- Microcontroller, Microprocessor and Its Interfaces-PIC, STM.
- Supports for the PCB layout design, Reviewing and Giving input to the Layout.
- Tools used for circuit design and simulation: Cadence Allegro, Proteus and LT Spice.
- Lab equipment used: Voltage/Current meter, LCR meter, DSO, Logic analyzer.
- Experience on Board Setup, Testing, Debugging and troubleshooting of PCB.
- Wireless Interfaces such as NRF (nRF24L01) Transceiver and Wi-Fi (ESP8266).

ROLE AND RESPONSIBILITIES

- Hardware life cycle management which includes Requirement collection, Architecture design, Circuit design, component selection, Schematic entry, Simulation, Prototype construction, Board and system level design, testing and debugging, Verification & Validation of a product.
- Designed circuits, selected components, created schematics, and supported PCB layout till Gerber release.
- Developed prototypes, conducted R&D, and tested hardware to meet customer specs.
- Supported microcontroller pin mapping and logic development.
- Created test plans, wrote test cases, and debugged hardware-software integration issues.
- Managed BOM and component procurement activities.

TOOLS AND IDEs

- Good experience on using OrCAD Capture CIS for Circuit designing.
- Good knowledge on using simulation tools like Proteus, LT spice.
- Agile methodology (JIRA software).

PROJECT EXCECUTED

SMART REMOTE FAN CONTROL: -

Project Summary: Designed a remote-control fan regulator using an IR remote and touch sensor for seamless operation.

Responsibilities:

- Designed hardware schematic with PIC12F1572.
- IR receiver, touch sensor, and relay driver.
- Conducted unpowered, impedance, and powered testing.
- Created schematics, test procedures, and user manuals.

SMART-HOME AUTOMATION: -

Project Summary: Smart switch Board with which user can control home appliances like Lights, fans and sockets connected to device via App (Wi-Fi), Remote(nRF) and manual Touch Switches.

Responsibilities: -

- Designed hardware architecture and block diagram as per project requirements.
- Designed and interfaced capacitive touch switches, Tri-color LEDs, and relay driver circuits.
- Integrated Wi-Fi (ESP) and nRF modules for app-based and remote control.
- Conducted bench-level testing and unit testing for each hardware module.
- Prepared design documents, test procedures, and test reports.
- Performed hardware debugging and resolved circuit-level issues.

REMOTE CONTROL PANEL: -

Project Summary: Developed a Remote-Control Panel with a microcontroller, LCD, and Switch Inputs to interface with an RHEL 8.1 workstation via RS-422 for two-way communication and control.

Responsibilities: -

- Developed hardware architecture, and circuit design based on the Requirement.
- Integrated microcontroller, LCD (16×2), keypad, and RS-422 driver.
- Implemented and tested RS-422 communication.
- Debugged hardware and documented design & test procedures.

EDUCATIONAL QUALIFICATION

Qualification	College/School	YOP	Per %
B.E(Electrical and Electronics Engineering)	KLS Vishwanthrao Deshpande Institute of technology, Haliyal, Karnataka	2023	8.5%
PUC	Aryabhata Pu science College, Dharwad, Karnataka	2019	82%
SSLC	Jaycee English Medium School Mahalingpur, Karnataka	2017	79%

PERSONEL INFORMATION

- Name : Rahul Khot
- Father Name : Channappa
- Date of Birth : 16-04-2001
- Gender : Male
- Marital Status : Unmarried
- Language Known : English, Kannada and Hindi.
- Hobbies : Learning New Skills, Political Podcast and Swimming.